

IAPI ARCHITECTURE STYLES

REST



Stateless architecture for scalable APIs using standard HTTP methods like GET, POST, PUT, and DELETE.



```
POST /api/message HTTP/1.1
Host: api.example.com
Content-Type: application/json

{
  "text": "Greetings, Earth!",
  "sender": "Alice123"
}
```

WEBSOCKET



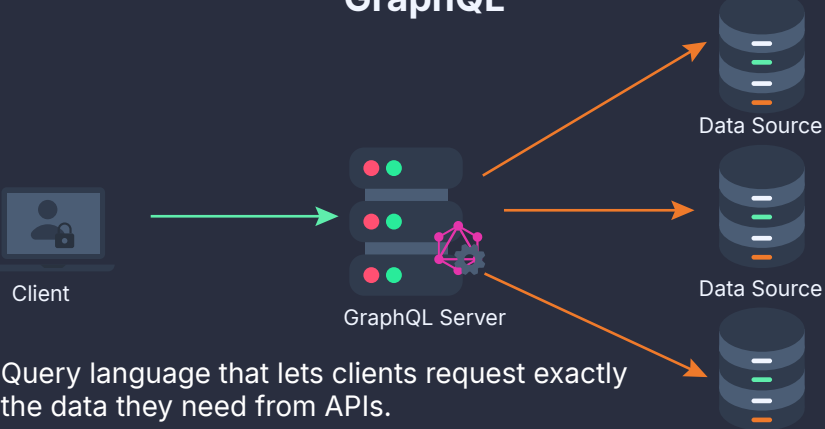
Enables full-duplex, real-time communication between client and server.



```
// JavaScript WebSocket client
const ws = new WebSocket("wss://example.com/socket");

ws.onmessage = (event) => {
  console.log("Message from server:", event.data);
};
ws.send("Hello Server!");
```

GraphQL

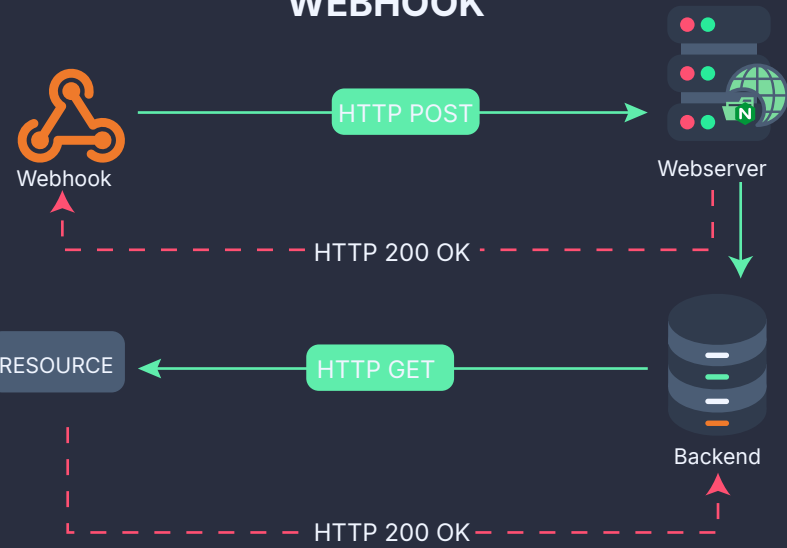


Query language that lets clients request exactly the data they need from APIs.



```
# GraphQL Query
{
  user(id: "123") {
    name
    email
  }
}
```

WEBHOOK



Server-side callback triggered by events; sends real-time HTTP POSTs to subscribed endpoints.

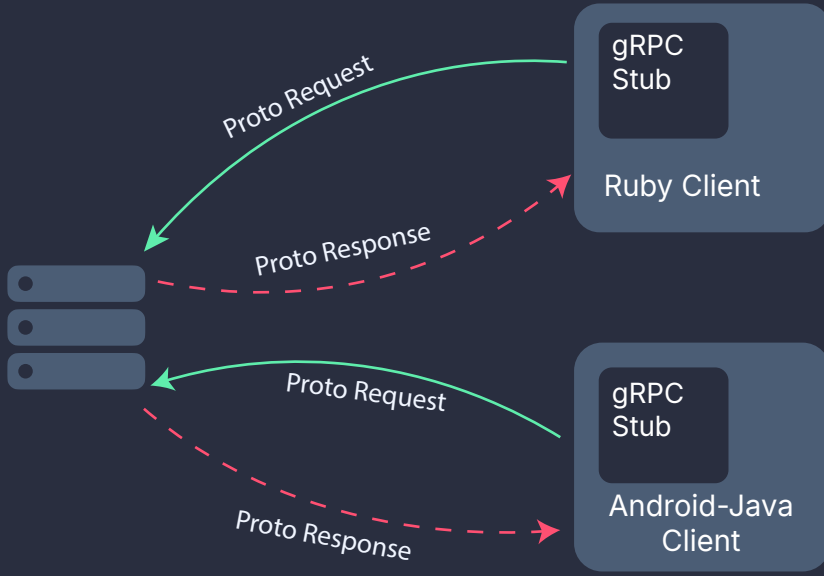


```
const webhookUrl = "https://webhook.site/webhook-id";

const data = {
  name: "brad",
  "Channel URL": "test url"
};

fetch(webhookUrl, {
  method: "POST",
  headers: {
    "Content-Type": "application/json"
  },
  body: JSON.stringify(data)
});
```

gRPC



High-performance RPC framework using Protocol Buffers for fast, compact data exchange.



```
// service.proto

service UserService {
  rpc GetUser (UserRequest) returns (UserResponse);
}

message UserRequest {
  int32 id = 1;
}

message UserResponse {
  string name = 1;
  string email = 2;
}
```

SOAP



XML-based protocol that runs over HTTP, SMTP, or other transport layers. Common in legacy enterprise systems.



```
POST /weather HTTP/1.1
Host: api.example.com
Content-Type: text/xml; charset=utf-8
SOAPAction: "getWeather"

<soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/">

  <soap:Body>

    <getWeather xmlns="http://example.com/">
      <city>Paris</city>
      <unit>Celsius</unit>
    </getWeather>
  </soap:Body>

</soap:Envelope>
```